

Question 12: PROGRAM TO PRINT ASCII CODE OF ANY CHARACTER.

```
#include<stdio.h>

int main(){

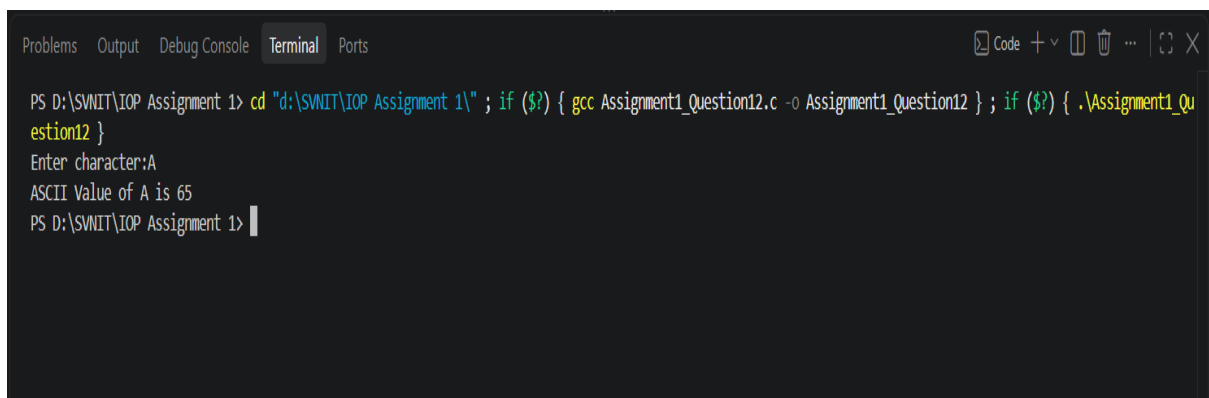
    char ch;

    printf("Enter character:");

    scanf("%c",&ch);

    printf("ASCII Value of %c is %d",ch,ch);

}
```



The screenshot shows a terminal window with the following content:

```
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question12.c -o Assignment1_Question12 } ; if ($?) { .\Assignment1_Question12 }
Enter character:A
ASCII Value of A is 65
PS D:\SVNIT\IOP Assignment 1>
```

The terminal window has tabs for Problems, Output, Debug Console, Terminal, and Ports. The Terminal tab is active, showing the command prompt and the execution of the program. The program prompts for a character, receives 'A', and prints its ASCII value as 65.

Question 13: PROGRAM TO CHECK WHETHER A NO. IS EVEN OR ODD

```
#include<stdio.h>

int main(){

    int n;

    printf("Enter a number:");

    scanf("%d",&n);

    if(n%2==0){

        printf("The number %d is Even",n);

    }

    else{

        printf("The number %d is Odd",n);

    }

}
```

PS D:\SVNIT\IOP Assignment 1>

```
> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question13.c -o Assignment1_Question13 } ; if ($?) { .\Assignment1_Qu
estion13 }
```

Enter a number:13

The number 13 is Odd

PS D:\SVNIT\IOP Assignment 1> █

Question 14: PROGRAM TO CHECK WHETHER A GIVEN CHARACTER IS CAPITAL, LETTER, SMALL CASE LETTER, A DIGIT OR A SPECIAL SYMBOL.

```
#include<stdio.h>

int main(){

    char ch;

    printf("Enter A CHARACTER:");

    scanf("%c",&ch);

    if(ch>=65&&ch<=90){

        printf("Character is a Upper case Letter");

    }

    else if(ch>=97&&ch<=122){

        printf("Character is a Lower case Letter");

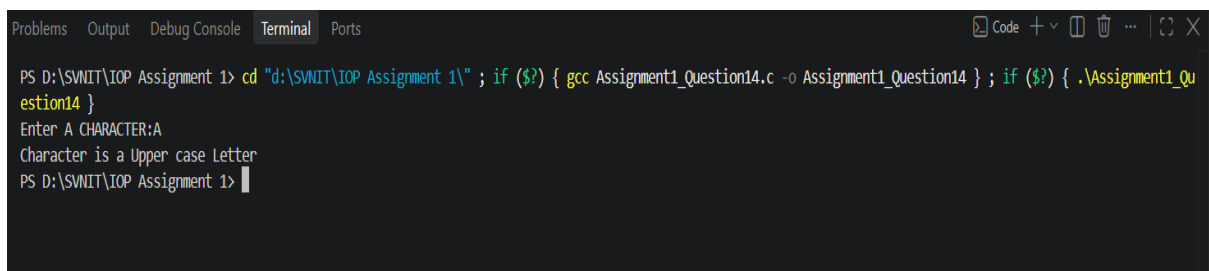
    }

    else{

        printf("Character is either a digit or special character");

    }

}
```



```
Problems Output Debug Console Terminal Ports
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question14.c -o Assignment1_Question14 } ; if ($?) { .\Assignment1_Question14 }
Enter A CHARACTER:A
Character is a Upper case Letter
PS D:\SVNIT\IOP Assignment 1> |
```

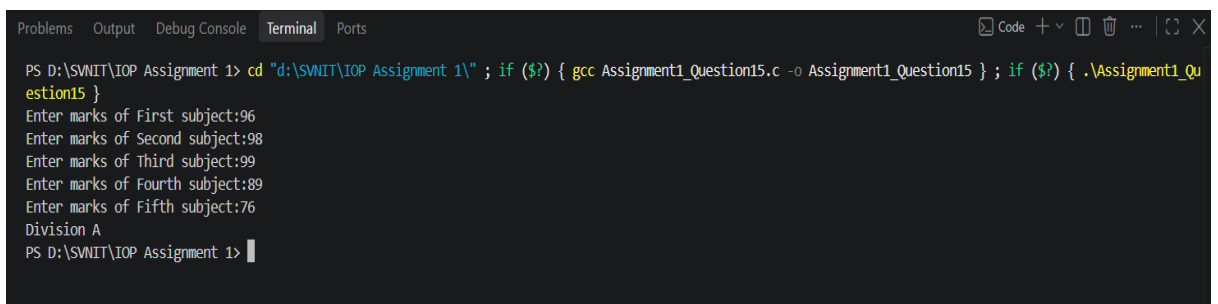
Question 15: PROGRAM TO READ MARKS OF FIVE SUBJECT AND PRINT DIVISION

```
#include<stdio.h>
int main(){
    int m1,m2,m3,m4,m5,total;
    float avg;
    printf("Enter marks of First subject:");
    scanf("%d",&m1);
    printf("Enter marks of Second subject:");
    scanf("%d",&m2);
    printf("Enter marks of Third subject:");
    scanf("%d",&m3);
    printf("Enter marks of Fourth subject:");
    scanf("%d",&m4);
    printf("Enter marks of Fifth subject:");
    scanf("%d",&m5);
    total=m1+m2+m3+m4+m5;
    avg=total/5;
    if(avg>=80){
        printf("Division A");

    }
    else if(avg>=60&&avg<=79){
        printf("Division B");

    }
    else if(avg>=40&&avg<=59){
        printf("Division c");

    }
    else{
        printf("Fail");
    }
}
```

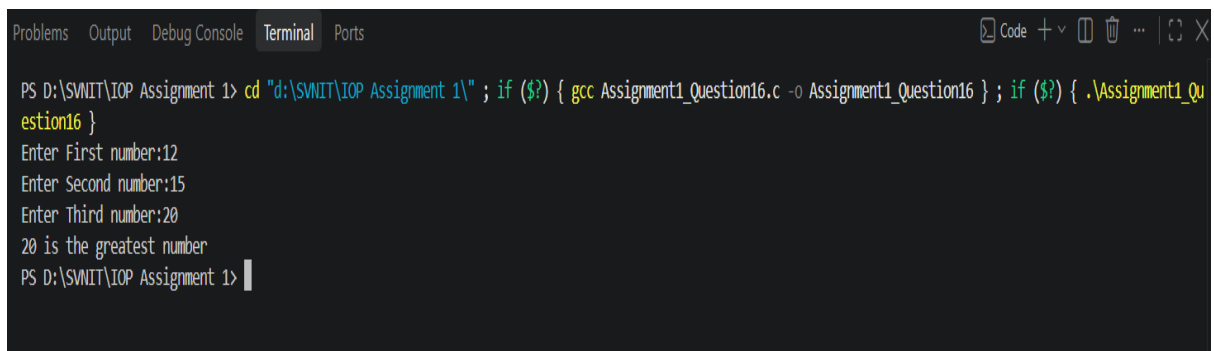


```
Problems Output Debug Console Terminal Ports
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question15.c -o Assignment1_Question15 } ; if ($?) { .\Assignment1_Question15 }
Enter marks of First subject:96
Enter marks of Second subject:98
Enter marks of Third subject:99
Enter marks of Fourth subject:89
Enter marks of Fifth subject:76
Division A
PS D:\SVNIT\IOP Assignment 1> |
```

Question 16: PROGRAM TO READ THREE NOS. AND PRINT MAX.

```
#include<stdio.h>
int main(){
    int a,b,c;
    printf("Enter First number:");
    scanf("%d",&a);
    printf("Enter Second number:");
    scanf("%d",&b);
    printf("Enter Third number:");
    scanf("%d",&c);
    if(a>b){
        if(a>c){
            printf("%d is the greatest number",a);

        }
        else{
            printf("%d is the greatest number",c);
        }
    }
    if(b>a){
        if(b>c){
            printf("%d is the greatest number",b);
        }
        else{
            printf("%d is the greatest number",c);
        }
    }
}
```



The screenshot shows a Windows Command Prompt window with the following content:

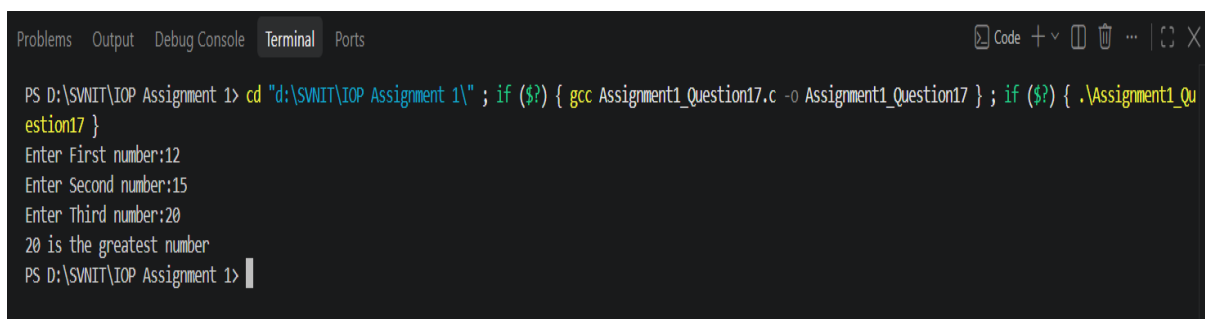
```
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question16.c -o Assignment1_Question16 } ; if ($?) { .\Assignment1_Question16 }
Enter First number:12
Enter Second number:15
Enter Third number:20
20 is the greatest number
PS D:\SVNIT\IOP Assignment 1>
```

The terminal window has tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active, showing the command prompt and the output of the program. The program correctly identifies 20 as the greatest number among 12, 15, and 20.

17: PROGRAM TO READ THREE NOS. AND PRINT MAX USING LOGICAL OPERATOR.(&&)

```
#include<stdio.h>
int main(){
    int a,b,c;
    printf("Enter First number:");
    scanf("%d",&a);
    printf("Enter Second number:");
    scanf("%d",&b);
    printf("Enter Third number:");
    scanf("%d",&c);
    if(a>b&&a>c){
        printf("%d is the greatest number",a);
    }
    else if(b>a&&b>c){
        printf("%d is the greatest number",c);
    }

    if(b>a&&b>c){
        printf("%d is the greatest number",b);
    }
    else if(c>a&&c>b){
        printf("%d is the greatest number",c);
    }
}
```

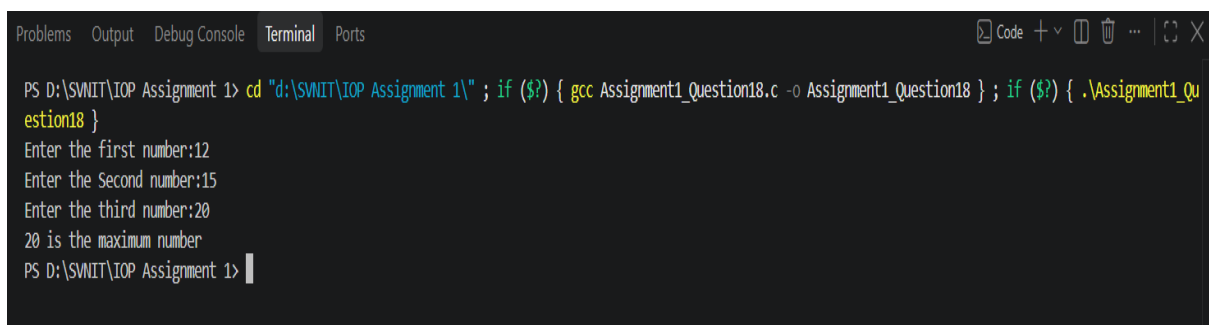


The screenshot shows a terminal window with a dark background. At the top, there are tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. The command prompt shows the user is in the directory 'D:\SWNIT\IOP Assignment 1'. The user enters three numbers: 12, 15, and 20. The program outputs '20 is the greatest number'.

```
PS D:\SWNIT\IOP Assignment 1> cd "d:\SWNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question17.c -o Assignment1_Question17 } ; if ($?) { .\Assignment1_Question17 }
Enter First number:12
Enter Second number:15
Enter Third number:20
20 is the greatest number
PS D:\SWNIT\IOP Assignment 1>
```

Question 18: PROGRAM TO READ THREE NOS. AND PRINT MAX USING CONDITIONAL (? :) OPERATOR

```
#include<stdio.h>
int main(){
    int n1,n2,n3;
    printf("Enter the first number:");
    scanf("%d",&n1);
    printf("Enter the Second number:");
    scanf("%d",&n2);
    printf("Enter the third number:");
    scanf("%d",&n3);
    int max = (n1 > n2 && n1 > n3) ? n1 : ((n2 > n3) ? n2 : n3);
    printf("%d is the maximum number",max);
}
```



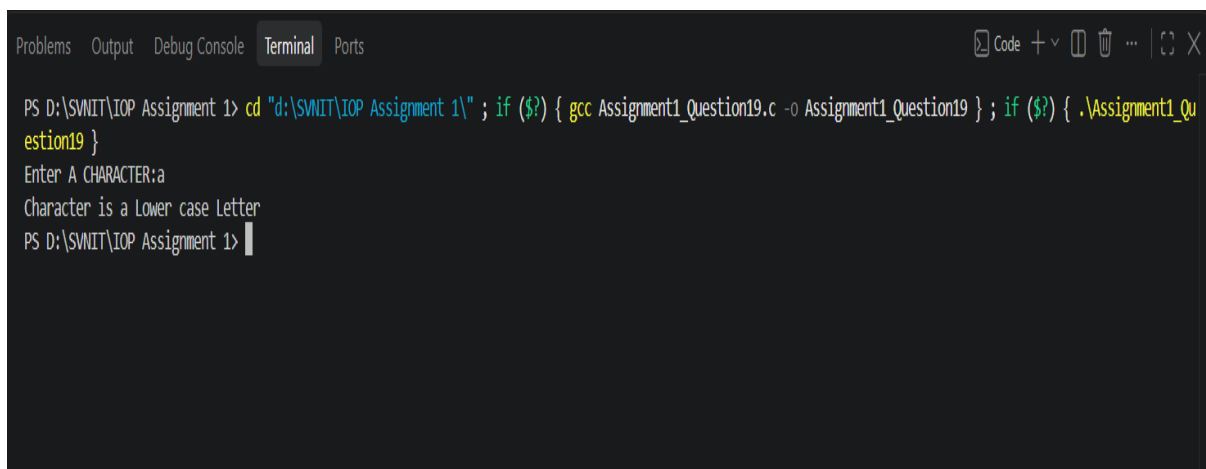
The screenshot shows a terminal window with the following content:

```
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question18.c -o Assignment1_Question18 } ; if ($?) { .\Assignment1_Question18 }
Enter the first number:12
Enter the Second number:15
Enter the third number:20
20 is the maximum number
PS D:\SVNIT\IOP Assignment 1> |
```

Question 19: PROGRAM TO READ A CHARACTER AND CHECK WHETHER IT IS A SMALL CASE LETTER OR NOT USING CONDATIONAL OPERATOR.

```
#include<stdio.h>
int main(){
    char ch;
    printf("Enter A CHARACTER:");
    scanf("%c",&ch);
    if(ch>=65&&ch<=90){
        printf("Character is not a lower case Letter");

    }
    else if(ch>=97&&ch<=122){
        printf("Character is a Lower case Letter");
    }
}
```

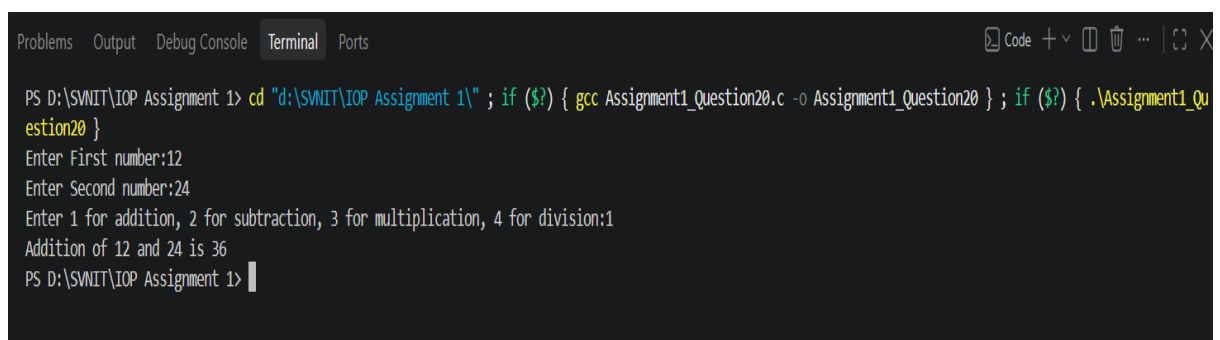


```
Problems Output Debug Console Terminal Ports
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question19.c -o Assignment1_Question19 } ; if ($?) { .\Assignment1_Question19 }
Enter A CHARACTER:a
Character is a Lower case Letter
PS D:\SVNIT\IOP Assignment 1>
```

Question 20: PROGRAM WHICH WILL WORK LIKE A SIMPLE CALCULATOR USING SWITCH-CASE.

```
#include<stdio.h>

int main(){
    int a,b;
    printf("Enter First number:");
    scanf("%d",&a);
    printf("Enter Second number:");
    scanf("%d",&b);
    int n;
    printf("Enter 1 for addition, 2 for subtraction, 3 for multiplication, 4 for division:");
    scanf("%d",&n);
    switch(n){
        case 1:
            printf("Addition of %d and %d is %d",a,b,a+b);
            break;
        case 2:
            printf("Subtraction of %d and %d is %d",a,b,a-b);
            break;
        case 3:
            printf("Multiplication of %d and %d is %d",a,b,a*b);
            break;
        case 4:
            if(b==0){//try of if in switch case
                printf("Division by zero is not allowed");
            }
            else{
                printf("Division of %d and %d is %f",a,b,(float)a/b);
            }
            break;
        default:
            printf("Invalid input");
    }
}
```



```
Problems Output Debug Console Terminal Ports
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question20.c -o Assignment1_Question20 } ; if ($?) { .\Assignment1_Question20 }
Enter First number:12
Enter Second number:24
Enter 1 for addition, 2 for subtraction, 3 for multiplication, 4 for division:1
Addition of 12 and 24 is 36
PS D:\SVNIT\IOP Assignment 1> 
```

Question 21: PROGRAM TO READ TWO NOS. AND PERFORM SPECIFIC TASK (USING ARITHMETIC OPERATOR)AND THIS PERFORM USING SWITCH-CASE.(CASE IN CHAR FORM).

```
#include<stdio.h>
int main(){
    int a,b;
    printf("Enter First number:");
    scanf("%d",&a);
    printf("Enter Second number:");
    scanf("%d",&b);
    char ch;
    printf("Enter A for addition, B for subtraction, C for multiplication, D for
division:");
    scanf("%c ", &ch);
    printf("You entered %c",ch);
    switch(ch){
        case 'A':
            printf("Addition of %d and %d is %d",a,b,a+b);
            break;
        case 'B':
            printf("Subtraction of %d and %d is %d",a,b,a-b);
            break;
        case 'C':
            printf("Multiplication of %d and %d is %d",a,b,a*b);
            break;
        case 'D':
            if(b==0){
                printf("Division by zero is not allowed");
            }
            else{
                printf("Division of %d and %d is %f",a,b,(float)a/b);
            }
            break;
        default:
            printf("Invalid input");
    }
}
```

}

```
c -o ASSIGNMENT_21 } ; if ($?) { .\ASSIGNMENT_21 }  
enter the 1st number : 4  
enter the 2nd number : 3  
enter A for Addition, B for Substraction, C for Multipication, D for Division : D  
DIVIDE=1.333333  
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB2>
```

Question 22: PROGRAM TO CALCULATE FACTORIAL OF A NO.

```
#include<stdio.h>

int main(){

    int a;

    printf("Enter the number of elements:");

    scanf("%d",&a);

    int fact=1;

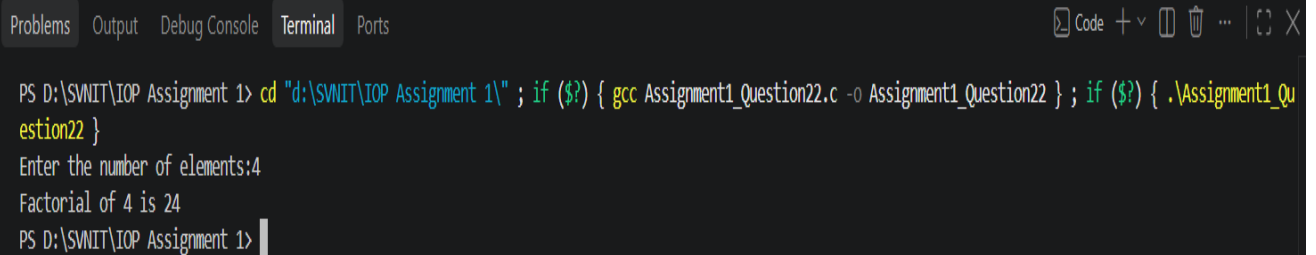
    for(int i=1;i<=a;i++){

        fact=fact*i;// Add code to read and process each element

    }

    printf("Factorial of %d is %d",a,fact);

}
```



The screenshot shows a terminal window with the following content:

```
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc Assignment1_Question22.c -o Assignment1_Question22 } ; if ($?) { .\Assignment1_Question22 }
Enter the number of elements:4
Factorial of 4 is 24
PS D:\SVNIT\IOP Assignment 1> |
```

The terminal window has tabs for Problems, Output, Debug Console, Terminal, and Ports. The Terminal tab is active, showing the command to compile and run the program. The output shows the program prompting for the number of elements, receiving 4, and calculating the factorial as 24.

Question 23: PROGRAM TO READ TWO NOS. AND CALCULATE POWER WITHOUT USING HEADER FILE().

```
#include<stdio.h>

#include<math.h>

int main(){

    int a,b;

    float power;

    printf("Enter two numbers: ");

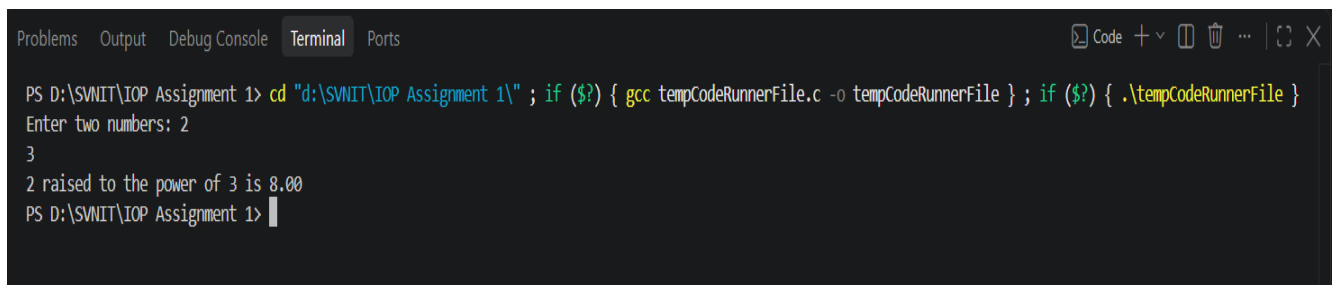
    scanf("%d %d",&a,&b);

    power=pow(a,b);

    printf("%d raised to the power of %d is %.2f",a,b,power);

    return 0;

}
```



```
Problems Output Debug Console Terminal Ports
PS D:\SVNIT\IOP Assignment 1> cd "d:\SVNIT\IOP Assignment 1\" ; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Enter two numbers: 2
3
2 raised to the power of 3 is 8.00
PS D:\SVNIT\IOP Assignment 1> |
```